

7x5 LED MATRIX Using Arduino

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Presented here is a simple 7x5 LED matrix circuit driven by Arduino Uno. It can be used to display your name or create any number of patterns. If the logic of this LED matrix is understood, it would be easy to handle much larger matrices and even make a 3D LED matrix.

Circuit and working

The circuit shown in Fig. 1 includes an Arduino UNO board, a few transistors that act as switches, a few resistors and LEDs. Arduino board controls the LEDs and their timings according to the code programmed into it. In the matrix arrangement, each LED can be addressed by specifying its location in terms of its row and column. For example, the

top-left LED is addressed as (0,0), the top-right as (0,6), the bottom-right one as (4,6) and so on.

Arduino Uno board. Arduino is an open source electronics prototyping platform based on flexible and user-friendly hardware and software. It can be used by anyone for creating interactive objects or environments. The board consists of an ATmega328 microcontroller (MCU), 14 digital I/O pins, six analogue inputs, a USB connection for programming the onboard MCU,

a power jack, an ICSP header and a reset button. It is operated using a 16MHz crystal oscillator. Arduino board is connected to the computer with a USB cable and programmed using Arduino IDE.

LED matrix. The LED matrix consists of 35 LEDs connected as per the circuit diagram. Seven transistors are used in the circuit to enhance switching action of the MCU. Collector terminal of each transistor (T1 through T7) is connected to 5V supply output from Arduino board.

Emitter terminals of the transistors are connected to each column of the matrix. Resistors R1 through R7 are connected to the bases of transistors and on their other sides to Arduino as row pins. Shorted pins of each column are connected

PARTS LIST

Semiconductors:

T1-T7 - BC547 npn transistor

LED1-LED35 - 5mm LED

Board1 - Arduino Uno board

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1-R7 - 1-kilo-ohm

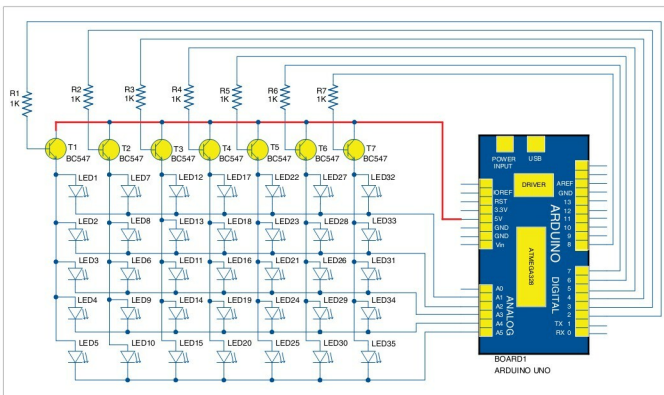


Fig. 1: Circuit diagram of the 7x5 LED matrix using Arduino